# Java Technical Interview Notes

Core Java, object-oriented programming, exceptions, and collections

## Java Fundamentals

- Java is a high-level, class-based programming language. Source code is compiled into bytecode that runs on a Java Virtual Machine (JVM).
- The JDK provides development tools. The JVM executes Java bytecode. JRE is a term for the runtime environment and its supporting libraries.
- Primitive types include byte, short, int, long, float, double, char, and boolean.
- Interview prompts: What are JDK, JRE, and JVM? What is bytecode?

## Object-Oriented Programming

- A class defines a type and behavior; an object is an instance of a class.
- Encapsulation combines data and methods and can restrict direct access using access modifiers.
- Inheritance allows a class to derive fields and methods from another class. Java classes use extends for class inheritance.
- Polymorphism lets one parent type or interface represent different implementations. Method overriding supports runtime polymorphism.
- Abstraction exposes essential behavior while hiding implementation details. Abstract classes and interfaces help implement abstraction.
- Interview prompts: Explain the four pillars of OOP. Compare method overloading and overriding.

## Strings and Equality

- String objects are immutable: operations that appear to modify a String produce another String value.
- The == operator compares primitive values or object references. equals() can compare logical content when implemented appropriately.
- StringBuilder supports mutable string operations and is useful when repeatedly building strings.
- Interview prompts: Why are Strings immutable? Compare == and equals().

## Exception Handling

- Java supports try, catch, finally, throw, and throws for exception handling.
- Checked exceptions are checked by the compiler. Unchecked exceptions include RuntimeException and its subclasses.
- A finally block is commonly used for cleanup, though it may not execute in every possible termination scenario.
- Interview prompts: Compare checked and unchecked exceptions. What is finally used for?

## Collections

- The Java Collections Framework provides interfaces and classes for groups of objects.
- List preserves positional order and permits duplicates. Set models unique elements. Map associates keys with values.
- ArrayList is a resizable array implementation. HashMap stores key-value mappings and does not guarantee iteration order.
- Interview prompts: Compare ArrayList and LinkedList. Compare List, Set, and Map.

## Multithreading and Coding Practice

- A thread is a path of execution within a process. Java provides threads and concurrency utilities.

- Synchronization coordinates access to shared mutable state. Poor synchronization can cause contention or deadlocks.
- Practice coding: reverse a string, check whether a number is prime, find the second-largest array element, count character frequencies, and remove duplicates.
- For each solution, explain the approach, time complexity, edge cases, and sample input/output.